

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION

Federal State Autonomous Educational
Institution of Higher Education "Southern Federal University"
(SOUTHERN FEDERAL UNIVERSITY)

COURSE SYLLABUS FOR
Higher Mathematics (Высшая математика)

Подписано электронной подписью:
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I. GOALS AND OBJECTIVES OF MASTERING THE DISCIPLINE

Objectives of mastering the discipline:

- formation of a mathematical foundation for the analysis, modeling and research of processes in the field of information technology and artificial intelligence;
- development of systems thinking and an interdisciplinary approach to solving professional problems in the field of IT and AI;
- mastering modern mathematical apparatus for setting and solving promising problems of artificial intelligence;
- Develop an understanding of basic mathematical methods and probabilistic modeling to work with data uncertainty.
- formation of skills in statistical data analysis and validation of AI models.

Objectives of mastering the discipline:

- study of the basic concepts and structures of various sections of higher mathematics;
- formation of ideas about mathematical modeling of processes and phenomena in information systems and artificial intelligence.
- mastering the principles of the Bayesian approach and building probabilistic models for data analysis and working with uncertainty;
- Study mathematical approaches in optimization methods for training and tuning machine learning models.
- acquisition of skills in statistical data analysis, hypothesis testing and quality assessment of AI models;
- development of the ability to integrate mathematical methods into engineering solutions for the tasks of the IT industry and artificial intelligence.

II. THE PLACE OF THE DISCIPLINE IN THE STRUCTURE OF THE EDUCATIONAL PROGRAM

The discipline belongs to the compulsory part of the educational program and is included in the module B1.M3.ED1 of elective disciplines.

This discipline is based on the basic knowledge, skills and abilities formed when obtaining the previous level of education.

Knowledge, skills and abilities formed by this discipline will be required when mastering the following elements of the educational program:

- "Generative Artificial Intelligence and Machine Learning"
- "Artificial Intelligence and Data Analysis";
- "Architecture, Technologies and Tools for the Development of AI-Enabled Software";
- "Algorithms, Technologies and Tools for Image and Video Sequence Processing"
- "Machine Learning in Computer Vision"
- "Optimization methods";
- "Numerical methods";
- "Mathematical Methods of Image Recognition";
- "Theory of Decision-Making";
- "Business Analytics";
- "Cryptographic methods of information protection";
- "Evolutionary modeling".

III. REQUIREMENTS FOR THE RESULTS OF MASTERING THE DISCIPLINE

The development of the discipline is aimed at the formation of the following competencies in accordance with the educational standard and educational program:

List of planned learning outcomes in the discipline correlated with indicators of competence achievement

Competence	Indicators of Competency Achievement	Learning outcomes
CPC-1 – Able to understand the natural science essence of problems that arise in the course of professional activity, apply interdisciplinary and general scientific knowledge and methods to solve problems of analysis, modeling and research of objects of professional activity	CPC-1.1 - Solves standard professional problems using natural science and general engineering knowledge, methods of mathematical analysis and modeling	Must know: - basic concepts of mathematics and mathematical methods of describing objects; - equations of lines, planes and simple surfaces; - parametric and polar coordinates; - basic operations with vectors and matrices; - linear transformations, determinants and eigenvalues; - convergence/divergence of sequences, determination of the limit and continuity of the function, connection between them; - the basics of differentiation and integration, their geometric meaning (in the context of the analysis of functions); - Conditions and methods for finding extremes of multiparameter functions; - definition of the gradient and its physical meaning; - Taylor's formula for approximating functions; - Sylvester's criterion: conditions of positivity and negativity of matrices; - least squares method for approximating data; - Determination and calculation of multiple integrals;
	CPC-1.2 - Conducts theoretical and experimental studies of objects of professional activity	Must be able to: - analyze functions: find extremes, analyze charts; - apply mathematical concepts to solving practical problems; - work with matrix methods to solve systems of equations - analyze multidimensional functions: find extremes and investigate properties; - work with matrices and determinants for the analysis of quadratic forms; Must be proficient in: - methods of solving problems using algorithms and mathematical analysis; - skills of working with data and their visualization in the context of mathematical models; - data analysis using the method of least squares.

Competence	Indicators of Competency Achievement	Learning outcomes
PC-1. MF-1 Able to apply modern theoretical mathematics to develop new algorithms and formulate promising AI problems	PC-1.2. MF-1.2. Applies the apparatus of probability theory, mathematical statistics and information theory to formulate and analyze artificial intelligence problems	Must know: - basic concepts and laws of probability theory, mathematical statistics and information theory; - principles of building probabilistic models and assessing their parameters; - methods for analyzing random processes and stochastic quantities used in artificial intelligence tasks; - modern approaches to the formalization of machine learning tasks using a probabilistic apparatus. Must be able to: - formulate artificial intelligence problems in probabilistic terms (distributions, conditional probabilities, Bayesian inference); - apply methods of mathematical statistics to analyze data, test hypotheses and evaluate model parameters; - use probabilistic and information methods to select and analyze machine learning algorithms; - assess the uncertainty and stability of AI models based on a probabilistic approach.
	PC-1.3. MF-1.3 Applies the modern mathematical apparatus of probability theory to study methods and models of machine learning	Must be proficient in: - skills in building and interpreting probabilistic and statistical data models; - tools for applying the apparatus of probability theory and information theory in the development and study of AI algorithms; - practical methods for integrating probabilistic methods into the process of designing and optimizing machine learning models.
PC-2. MF-2 Able to apply a Bayesian approach to build probabilistic models, analyze uncertainty, and build adaptive AI systems	PC-2.1 MF-2.1 Uses Bayes' Theorem and its consequences, understands the differences between Bayesian statistics and the frequency approach	Must know: - the main provisions of Bayes' theorem and its consequences; - Differences between Bayesian and frequency statistics, their areas of application; - methods of a priori and a posteriori distribution, the concepts of likelihood and a priori assumptions; - principles of building Bayesian models for regression, classification and clustering problems; - Methods of approximation and computational procedures in Bayesian inference; - the role of Bayesian methods in the analysis of uncertainty and the creation of adaptive AI systems. Must be able to:

Competence	Indicators of Competency Achievement	Learning outcomes
	PC-2.2 MF-2.2 Applies Bayesian estimation methods and Bayesian intervals to solve statistical problems and build models	- Apply Bayes' Theorem to recalculate probabilities and update knowledge when new data arrives. - formulate and solve problems of estimating parameters using Bayesian methods and constructing confidence (a posteriori) intervals; - Develop and analyze Bayesian models for regression, classification, and clustering - Conduct uncertainty analysis and interpretation of results based on posteriori distributions; - Select and configure appropriate priors for specific AI tasks. Must be proficient in: - skills in the practical construction and application of Bayesian models for various types of data;
	PC-2.3 MF-2.4 - Applies Bayesian models to build and analyze regression, classification, clustering, improving the stability and interpretability of modern AI models	- the ability to integrate the Bayesian approach into the process of developing adaptive systems and models of artificial intelligence; - techniques for visualizing and interpreting posteriori distributions to increase the transparency and stability of decisions.
SC 3. MF-3 Capable of applying modern optimization techniques to train machine learning models, tune hyperparameters, and solve artificial intelligence problems	SC 3.1 MF-3.1 Applies optimization techniques to the development and research of training algorithms	Must know: - Fundamental optimization concepts: function extremum, first- and second-order conditions, convexity, and convergence; - analytical methods for finding stationary points and studying them to a minimum/maximum; - the main classes of optimization methods: gradient and stochastic methods, Lagrange multiplier methods for problems with constraints; - the role of optimization methods in training machine learning models and tuning hyperparameters; - Mathematical foundations of convergence and stability of optimization algorithms. Must be able to: - formulate optimization problems in an analytical form (objective function, constraints, optimality conditions); - to determine and classify the types of extreme problems (conditional, unconditional, convex, non-convex); - perform analytical solutions of the simplest optimization problems by means of differential and integral calculus; - apply Lagrange methods to find a conditional extremum; - Interpret the mathematical results of optimization in the context of machine learning and artificial intelligence problems. Must be proficient in:

Competence	Indicators of Competency Achievement	Learning outcomes
		- skills of formalizing AI tasks in mathematical form; - methods of mathematical analysis for the study of extremums of functions of one and several variables; - skills in building and researching optimization models; - the methodology for choosing an adequate approach to optimization depending on the properties of the objective function and constraints; - a mathematical apparatus for assessing the stability and correctness of the solutions found.
PC-4. MF-4 Able to apply statistical techniques to data analysis, validate machine learning models, and conduct AI experiments	PC-4.1 MF-4.1 Applies statistical methods of analysis and machine learning to solve data analysis problems and conduct experiments on data	Must know: - methods of data preparation and pre-processing; - Basic concepts of mathematical statistics: sampling, general population, distribution parameters, confidence intervals, hypothesis testing; - types of probability distributions and their characteristics (mathematical expectation, variance, correlation); - principles of statistical inference and evaluation of parameters; - Fundamentals of analysis of variance, correlation and regression analysis; - the role of statistical methods in data analysis, validation of machine learning models, and design of experiments in AI. Must be able to: - formulate data analysis tasks in terms of mathematical statistics; - select and substantiate statistical criteria for testing hypotheses; - Calculate and interpret sample characteristics and confidence intervals - apply methods of correlation and regression analysis to describe the relationships between variables; - evaluate the reliability and significance of the results of experiments without the use of software. Must be proficient in: - skills of analytical solution of typical statistical problems by means of higher mathematics; - methods of planning and mathematical substantiation of experiments with data; - approaches to the interpretation of statistical results for the analysis and validation of machine learning models; - a mathematical apparatus for substantiating conclusions about patterns and relationships in data.

IV. CONTENT AND STRUCTURE OF THE DISCIPLINE

The complexity of the discipline is 10 credits, 360 hours,
including 1 credit, 36 hours for the exam

Form of intermediate attestation: differential credit (1st semester), exam (2nd semester)

4.1. The content of the discipline, structured by the topics of classes, 1 semester

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)				Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson	
		Contact work								Independe nt work
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
Module 1. Linear and Vector Algebra, Analytic Geometry										
1	Topic 1. The concept of matrix and matrix determinant. SLOUGH 1.1. Matrices, actions on matrices 1.2. Matrix determinant, calculation of determinants, properties of determinants 1.3. Inverse matrix, matrix view of SLOUGH 1.4. Matrix method for solving SLOUGH 1.5 Storage and processing of data in the form of matrices	2	-	–	-	2	Informational lecture	-	-	-
2	Calculate the determinant of the matrix. Actions on matrices.	-	2	-	-	2	Seminar	-	-	-
3	Inverse matrix. Application of the matrix method of solving SLOUGH	-	2	-	-	2	Seminar	-	-	-
4	Topic 2. Some methods for solving SLOUGH. Matrix rank and its properties 2.1. Kramer's method of solving SLOUGH 2.2. Gaussian method for solving SLOUGH 2.3. Matrix Rank and Its Properties 2.4. Kronecker-Capelli theorem 2.5 Use of SLAS in determining the relationship between input data and result	2	-	-	-	2	Informational lecture	-	-	-
5	Application of the Kramer method in the calculation of SLOUGH. Application of the Gaussian method in the calculation of SLOUGH		2			2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
6	Topic 3. Vector algebra, basis and coordinates 3.1. Geometric vector, linear operations on vectors 3.2. Linear dependence and independence of vectors 3.3. Cartesian coordinate system on the plane and in space 3.4. Coordinate Transformation in Linear Operations on Vectors 3.5 Data as Feature Vectors	2	-	-	-	2	Informational lecture	-	-	-
7	Elements of linear and vector algebra. Determination of linear dependence/independence of vectors.	-	2	-	-	2	Seminar	-	-	-
8	Operations on vectors.	-	2	-	-	2	Seminar	-	-	-
9	Topic 4. Scalar, vector, and mixed product of vectors 4.1. Dot product of vectors 4.3. Vector product of vectors 4.3. Mixed product of vectors 4.4. Computation of a mixed product in the Cartesian coordinate system 4.5 Basic Affinity Metric in ML	2	-	-	-	2	nformational lecture	-	-	-
10	Computation of dot, vector, and mixed product in DSC. Linear dependence. Fundamental Decision System	-	2	-	-	2	Seminar	-	-	-
11	Topic 5. Linear transformations (LP), eigenvectors, and eigenvalues. Singular decomposition. Straight line and plane 5.1. LP of vectors on the plane and in space 5.2. Eigenvectors and eigenvalues 5.3. Singular expansions. Principal Component Method 5.4. Equation of a line on a plane and in space 5.5. Equation of a plane in space	2	-	-	-	2	nformational lecture	-	-	-
12	Finding eigenvectors and eigenvalues of LPs. Equations of plane and line in space.	-	2	-	-	2	Seminar	-	-	-
13	Elements of algebra and analytic geometry. Lesson with test work	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
14	Topic 6. Curves and 2nd Order Surfaces 6.1. 2nd order curves on a plane 6.2. Ellipse 6.3. Hyperbole 6.4. Parabola 6.5. Surfaces of the 2nd order. Geometric Shapes of Curves and Surfaces in Data Analysis	2	-	-	-	2	Informational lecture	-	-	-
15	Analytical geometry elements. Defining the Curve Type and Casting to Canonical View	-	2	-	-	2	Seminar	Test No1	Counter. Work	30
Module 2. Mathematical Analysis										
16	Topic 7. Properties of Numeric Sets and the Limit of Numerical Sequence 7.1. Numerical sets. Faces of numeric sets in ML: definition of valid feature ranges, normalization and standardization of data 7.2. Numerical sequence. Uniqueness of the limit and limitation of the converging numerical sequence in the analysis of successive approximations in model training, guarantee of convergence of optimization algorithms 7.3. Infinitesimal Quantities and Their Properties in Understanding Small Changes in Parameters in Neural Network Training 7.4. Newton's binomial. Number e 7.5. The principle of nested segments 7.6. Bolzano-Weierstrass theorem - understanding the stability of learning. 7.7. Cauchy Criterion and Sequence Convergence Verification in Model Training	2	-	-	-	2	Informational lecture	-	-	-
17	Applying the properties of numeric sets, calculating the limit of a numerical sequence.	-	2	-	-	2	Seminar	-	-	-
18	Application of properties of infinitesimal quantities. Properties of limits and monotonic sequences	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
19	Topic 8. The limit of the function and its properties 8.1. The limit of the function (Cauchy and Heine) is the understanding of the convergence of algorithms. 8.2. Small and large values of functions — the stability of models under extreme data. 8.3. Properties of limits — approximation and decomposition for optimization. 8.4. Remarkable Limits 8.5. Comparison of small values — the speed of convergence of teaching methods. 8.6. Continuity and Discontinuities of Functions – Choice of Activations and Stability of Learning. 8.7. Continuity of elementary functions — the correctness of the use of log, exp and power functions. 8.8. Continuity on an interval — the existence of minima and the convergence of optimization.	2	-	-	-	2	Informational lecture	-	-	-
20	Calculate the limit of a function. Uncovering uncertainties using limit properties. Calculate the limit of a function. Continuity	-	2	-	-	2	Seminar	-	-	-
21	Topic 9. Function differential, derivative. Calculating Derivatives and Differentials of a Function 9.1. Increment of a function and its argument, approximation of functions. Differential and Derived Function Definition 9.2. Geometric and mechanical meaning of the derived function; Error Change Rate 9.3. Rules for calculating a derived function. Table of Derivatives of Elementary Functions 9.4. Derivatives and Differentials of Higher Orders 9.5. Derivatives and Differentials of Functions Specified Parametrically 9.6. Implicit Function Derivatives and Optimization in Tasks with Constraints and Hidden Dependencies	2	-	-	-	2	Informational lecture	-	-	-
22	Computation of derivatives and differentials of elementary functions	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
23	Calculation of derivatives and differentials of higher-order functions specified parametrically and implicitly	-	2	-	-	2	Seminar	-	-	-
24	Topic 10. Basic theorems of differential calculus. 10.1. Increasing, decreasing, and extremes of functions—finding the minima of loss functions when training models. 10.2. Extremum conditions (Fermat's theorem) are criteria for finding optimum points in algorithms. 10.3. Rolle, Lagrange, and Cauchy theorems — substantiation of the existence of optimal points and the correctness of optimization methods.	2	-	-	-	2	Informational lecture	-	-	-
25	Study of the function for monotonicity, finding the extremum points of the function. Methods of differentiation	-	2	-	-	2	Seminar	-	-	-
26	Topic 11. L'Hôpital's rule. Taylor's formula. Examining a Function and Plotting It 11.1. L'Hôpital's Rule — Simplification of the Analysis of the Limiting Behavior of the Loss and Activation Functions. 11.2. The Taylor series and remainder estimation is an approximation of functions used in model optimization and analysis. 11.3. Monotonicity and extremum conditions — determining the decreasing directions of the error and local lows. 11.4. Convexity, Concavity, and Inflection Points — Analysis of the Terrain of the Loss and Convergence Functions of Algorithms. 11.5. Asymptotes of the function graph — the behavior of models at large feature values. 11.6. Study and graphing of functions — visualization of loss functions, activations, and learning dynamics.	2	-	-	-	2	Informational lecture	-	-	-
27	Application of L'Hôpital's rule. Decomposition of a function using the Taylor formula	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
28	Introduction to the analysis. Lesson with test work	-	2	-	-	2	Seminar	Test No2	Counter. Work	35
29	Topic 12. Primitive, indefinite integral. Methods for calculating an indeterminate integral 12.1. Primitive and indefinite integrals—application in the analysis of loss and activation functions. 12.2. Basic Integrals — the use of standard formulas in the analytical calculation of models. 12.3. Integration rules — simplification of expressions when displaying algorithm properties. 12.4. Piecemeal integration — decomposition of complex functions in gradient analysis. 12.5. Bringing under the differential sign — working with integrals that depend on the parameters of the model. 12.6. Variable substitution — data transformation and simplification of integrals in probabilistic models.	2	-	-	-	2	Informational lecture	-	-	-
30	Application of methods for calculating an indefinite integral and the simplest methods of integration. Application of the integration method in parts. Application of Variable Substitution and Integral Sign Reduction Methods	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independen t work				
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
31	Topic 13. Replacement of the variable and bringing under the differential sign. Integration of rational, trigonometric and irrational expressions 13.1. Integrals of rational functions — analysis and simplification of expressions in models. 13.2. Denominator factorization — preparation of functions for analytical calculations in probabilistic models. 13.3. Decomposition of fractions into protozoa — simplification of integrals in the derivation of distributions. 13.4. Trigonometric substitutions are work with periodic functions and features in models. 13.5. Integrals of irrational expressions are an extension of the set of functions for the analysis of complex models.	2	-	-	-	2	Informational lecture	-	-	-
32	Integration of rational functions. Method of representing a regular fraction as a sum of the simplest fractions and integration	-	2	-	-	2	Seminar	-	-	-
33	Integration of trigonometric and irrational expressions. Universal trigonometric substitution	-	2	-	-	2	Seminar	-	-	-
34	14. Definite integral, properties and computation. Improper Integrals 14.1. A definite integral and its properties are the calculation of areas, probabilities, and normalizations of distributions. 14.2. Integral with a variable upper limit. 14.3. Newton–Leibniz formula.					2				
35	Calculating a Definite Integral, Applying Properties	-	2	-	-	2	Seminar	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independe nt work				
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
36	Topic 15. FNP, limit continuity, partial derivatives and differentials 15.1. Euclidean space and FMP - multidimensional features, data vectors. 15.2. Limit and continuity of FMP - convergence of functions, stability of models. 15.3. Partial Derivative and Full Differential 15.4. Equations of plane and tangent 15.5. Differentiation of complex FNP - chain rule, backpropagation in neural networks. 15.6. Partial derivatives and differentials of higher orders - Hessian, accelerated optimization. 15.7. The Mixed Derivatives Theorem - Symmetry of Second Derivatives, Simplification of Calculations.	2	-	-	-	2	Informational lecture	-	-	-
37	Integrals of FOP, their calculation. Lesson with test work	-	2	-	-	2	Seminar	Test No3	Counter. Work	35
38	Private derivatives and differentials of FNP. Tangent and Surface Normal Equations. Differentiation of complex FNP.	-	2	-	-	2	Seminar	-	-	-
39	Topic 16. Taylor's formula, FNP extreme. Quadratic matrix, Sylvester's criterion. Hessian and its application for the analysis of the curvature of a function 16.1. Taylor's formula for FMP - local approximation of loss functions, estimation of model behavior. 16.2. Extremum points, a necessary condition is the search for candidates for the minimum/maximum of the loss function. 16.3. Extreme, sufficient condition - checking whether the point is a minimum/maximum, confidence in optimization. 16.4. Second Differential Matrix - Curvature Analysis of the Loss Function for Accelerated Optimization. 16.5. Sylvester's criterion in the classification of extremes; stability of the solution.	2	-	-	-	2	Informational lecture	-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
40	Partial derivatives and differentials of higher orders. Applying the Taylor formula and calculating the FNP extremum. Hessian and Function Curvature Analysis	-	2	-	-	2	Seminar	-	-	-
41	Topic 17. Method of least squares, conditional extremum 17.1. Least squares method - training of regression models, minimization of prediction error. 17.2. Conditional extremum of FNP - optimization with constraints, regularization, training with constraints. 17.3. Derivative by direction, gradient FNP - direction of the greatest growth of the loss function, gradient descent.	2	-	-	-	2	Informational lecture	-	-	-
42	Method of least squares. Conditional extremum of the FNP	-	2	-	-	2	Seminar	-	-	-
43	Calculation of a derivative by direction, gradient	-	2	-	-	2	Seminar	-	-	-
44	Topic 18. Double Integral, Its Properties, and Computation 18.1. Problems leading to a double integral, definition – calculation of areas and masses in multidimensional data, probability density integrals. 18.2. Dual Integral Properties – Linearity and Partitioning of Domains, Simplification of Multivariate Expectation Calculations. 18.3. Reduction of the double integral to the repeated one – numerical integration along the axes, estimation of multidimensional sums. 18.4. A general type of variable replacement in a double integral is to simplify integration, transform data for densities. 18.5. Polar coordinate system for double integral – integration in symmetric domains, processing of circular and radial data.	2				2		-	-	-
45	Calculation of a double integral, application of its properties. Application of the general form of variable substitution in a double integral		2			2		-	-	-

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)				Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson	
		Contact work								Independe nt work
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
Total Hours: 180		36	54		-	90		—		100

4.2. The content of the discipline, structured by the topics of classes, semester 2

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)				Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson	
		Contact work								Independe nt work
		Lectures	Prakt. Classes	Lab. Classes	Consultatio ns					
Module 1. Probability Theory and Random Variable Theory										
1	Topic 1. Probability. Probability theorems 1.1. Classical, geometric, statistical probability 1.2. Space of events. Axiomatika TV 1.3. Theorems of addition and multiplication 1.4. Full probability formula. Conditional probability 1.5 Bayes' formula. Priors and posteriori probabilities	2	-	-	-	-	Informational lecture	-	-	-
2	Combinatorial formulas and rules. Classical and geometric probability	-	2	-	-	2	Seminar	-	-	-
3	Theorems of addition and multiplication	-	2	-	-	2	Seminar	-	-	-
4	Topic 2. Discrete and continuous random variables 2.1. DSV and ways to describe it 2.2. DSV and ways to describe it 2.3. Distribution function and its density 2.4. Geometric interpretation of RF and PR	2	-	-	-	-	Informational lecture	-	-	-
5	Bayes' Formula	-	2	-	-	2	Seminar	-	--	
6	Topic 3. Numerical Characteristics of Discrete and Continuous Random Variables 3.1. Expectation of Mathematics 3.2. Dispersion and RMS 3.3. Mode and median 3.4. Moments of higher orders 3.5 Asymmetry and excess	2	-	-	-	-	Informational lecture	-	-	-
7	Bernoulli's formula and its approximations (Laplace's and Poisson's theorems)	-	2	-	-	2	Seminar	-	-	-
8	Test No1	-	2	-	-	2	Seminar	Test No1	Counter. work.	10

No p/n	Topic of the lesson	Types of educational work and their labor intensity, hours (including the use of online courses)					Form of the lesson	Name of the valuation	Type of control	Sum of points per lesson
		Contact work				Independent work				
		Lectures	Prakt. Classes	Lab. Classes	Consultations					
9	Topic 4. Basic Laws of Random Variable Distribution 4.1. Uniform, geometric, hypergeometric distributions of the DSV. Bernoulli distribution 4.3. Uniform, linear, normal, exponential distribution of NSV. 4.3. Lognormal distribution. Student's distributions and chi-square	2	-	-	-	-	Informational lecture	-	-	-
10	Table and Function of Distribution of Voluntary Resources. Numerical characteristics of the DSV: expected value, mode, median, variance, deviation, asymmetry, excess, moments	-	2	-	-	2	Seminar	-	-	-
11	Topic 5. Random vectors 5.1. Multidimensional SV 5.2. Distribution function and density of a multidimensional random vector 5.3. Conditional Distributions of Dry Matter 5.4. Basic Distributions of Land Plots	2	-	-	-	-	Informational lecture	-	-	-
12	Distribution function and density of NSV. Numerical characteristics of the NSV: expected value, mode, median, variance, deviation, asymmetry, kurtosis, moments	-	2	-	-	2	Seminar	-	-	-
13	Basic Laws of Random Variable Distribution	-	2	-	-	2	Seminar	-	-	-
14	Topic 6. Numerical characteristics of random vectors. Correlation 6.1. Random Vector Moments 6.2. Conditional expectation of the SV 6.3. Correlation coefficient of the DSV and the NSV. Coefficient of Determination	2	-	-	-	-	Informational lecture	-	-	-
15	Two-dimensional random variables: description, distribution function, density, conditional density	-	2	-	-	2	Seminar	-	-	-

Module 2. Mathematical statistics and regression analysis										
16	Topic 7. Sampling. Estimating Sample Distributions 7.1. Sampling and its properties. Small Samples 7.2. Statistical and grouped statistical series 7.3. Relationship of frequencies with probability 7.4. Evaluation of distributions	2	-	-	-	-	Informational lecture	-	-	-
17	Numerical Characteristics of Random Vectors	-	2	-	-	2	Seminar	-	-	-
18	Correlation of DSV and NSV. Determination.	-	2	-	-	2	Seminar	-	-	-
19	Topic 8. Point Estimates of Distribution Parameters 8.1. Point estimates of distribution parameters 8.2. Moment Method 8.3. Method of Greatest Likelihood	2	-	-	-	-	Informational lecture	-	-	-
20	Test No2	-	2	-	-	2	Seminar	Test No2	Counter. Work	15
21	Topic 9. Interval estimates of distribution parameters 9.1. Interval estimates of distribution parameters. Confidence interval 9.2. Statistical Testing of Hypotheses 9.3. Errors of the 1st and 2nd types	2	-	-	-	-	Informational lecture	-	-	-
22	Samples, their types and properties. Point Distribution Estimates	-	2	-	-	2	Seminar	-	-	-
23	Most Likely Method	-	2	-	-	2	Seminar	-	-	-
24	Topic 10. Parametric and nonparametric criteria 10.1. Parametric and Nonparametric Criteria 10.2. Pearson, Student, Fisher, Romanovsky criteria 10.3. Mann-Whitney, Wilcoxon, Kolmogorov criteria 10.4. Relationship between confidence intervals and statistical criteria	2	-	-	-	-	Informational lecture	-	-	-
25	Interval Estimates of Distribution	-	2	-	-	2	Seminar	-	-	-

26	Topic 11. Regression analysis. Evaluating Statistical Dependencies 11.1. Correlation Analysis 11.2. Regression Analysis 11.3. OLS 11.4. Regression curves 11.5. Estimating Statistical Dependencies	2	-	-	-	-	Informational lecture	-	-	-
27	Criteria for consent. Pearson's criterion	-	2	-	-	2	Seminar	-	-	-
28	Parametric and nonparametric criteria	-	2	-	-	2	Seminar	-	-	-
29	Topic 12. Bayesian regression 12.1. Bayesian regression 12.2. Differences from classical regression	2	-	-	-	-	Informational lecture	-	-	-
30	Regression, types of regression, OLS	-	2	-	-	2	Seminar	-	-	-
31	Topic 13. Multivariate Probabilistic Models and Stochastic Processes 13.1. Multivariate Probabilistic Models 13.2. Using MVM in Machine Learning 13.3. Stochastic Processes, Their Types and Characteristics	2	-	-	-	-	Informational lecture	-	-	-
32	Interval estimates for regression. Adequacy of the model	-	2	-	-	2	Seminar	-	-	-
33	Bayesian regression	-	2	-	-	2	Seminar	The defense of the indivs. Tasks No1	Prakt. Task	20
34	Topic 14. Markov Processes 14.1. The Markov Process 14.2. Markov chain, transition matrix 14.3. Hidden Markov Processes 14.2. The Use of MT in Neural Network Training Models	2	-	-	-	-	Informational lecture	-	-	-
35	Multivariate Probabilistic Models	-	2	-	-	2	Seminar	-	-	-

36	Topic 15. Time series 15.1. Time Series 15.2. Autoregression 15.3. Multidimensional time series 15.4. Time Series Applications: Gradient Boosting, Random Forest, Recurrent and Convolutional Networks, Transformers	2	-	-	-	-	Informational lecture	-	-	-
37	Random processes	-	2	-	-	2	Seminar	-	-	-
38	Markov chains	-	2	-	-	2	Seminar	-	-	-
39	Topic 16. Bayesian modeling 16.1. Bayesian modeling 16.2. Bayesian Networks	2	-	-	-	-	Informational lecture	-	-	-
40	Time series	-	2	-	-	2	Seminar	-	-	-
41	Topic 17. The Law of Large Numbers 17.1. ZBC 17.2. Consequences from the BCA 17.3. LCH and Statistics	2	-	-	-	-	Informational lecture	-	-	-
42	The LCH and its consequences	-	2	-	-	2	Seminar	-	-	-
43	Defense of an individual task	-	2	-	-	2	Seminar	-	-	-
44	Topic 18. Review: statistical methods of data analysis, experimentation 18.1. Review of statistical methods for data analysis	2	-	-	-	-	Informational lecture	-	-	-
45	Defense of an individual task	-	2	-	-	2	Seminar	The defense of the indivs. Tasks No2	Prakt. Task	15
Intermediate certification		—		—	—	36	Exam	Copies. Semester 3 Questions & Tickets	Exam	40
Total Hours: 180		36	54		-	90		—		100

4.3. Plan of extracurricular independent work, semester 1

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Linear and Vector Algebra, Analytic Geometry						
1	Topic 1. The concept of matrix and matrix determinant. SLOUGH	1	1	- Study and repetition of the material of lectures and practical classes	4	[2], [6] - [11], additional [16] - [20]
2	Topic 2. Some methods for solving SLAI. Matrix Rank and Its Properties	1	2	- Study and repetition of the material of lectures and practical classes	4	[2], [6] - [11], additional [16] - [20]
3	Topic 3. Vector algebra, basis and coordinates	1	3	- Study and repetition of the material of lectures and practical classes	4	[2], [6] - [11], additional [16] - [20]
4	Topic 4. Vectors, linear operations on vectors. Linear dependence and independence of vectors. Coordinate Transformation for Linear Operations on Vectors	1	4	- Study and repetition of the material of lectures and practical classes	4	[2], [6] - [11], additional [17], [25]
5	Topic 5. Linear transformations (LP), eigenvectors, and eigenvalues. Singular decomposition. Straight line and plane	1	5	- Study and repetition of the material of lectures and practical classes	4	[1], [5], [6], [7], additional [4], [5], [6]
6	Topic 6. Curves and 2nd Order Surfaces	1	6	- Study and repetition of the material of lectures and practical classes - Preparation for the test	4	[2], [6], [8] - [11], additional [17], [24]
Module 2. Mathematical Analysis						
7	Topic 7. Properties of Numeric Sets and the Limit of Numerical Sequence	1	7	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8], [11], additional [16] - [18]
8	Topic 8. The limit of the function and its properties	1	8	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8], [11], additional [16] - [18]
9	Topic 9. Function differential, derivative. Calculating Derivatives and Differentials of a Function	1	9	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8], [11], additional [16] - [18]
10	Topic 10. Basic theorems of differential calculus.	1	10	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8], [11], additional [16] - [18]
11	Topic 11. L'Hôpital's rule. Taylor's formula. Examining a Function and Plotting It	1	11	- study and repetition of the material of lectures and practical classes; - Preparation for the test	4	[2], [6], [8] - [11], additional [16] - [19]
12	Topic 12. Primitive, indefinite integral. Methods for calculating an indeterminate integral	1	12	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8] - [11], additional [16] - [19]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
13	Topic 13. Replacement of the variable and bringing under the differential sign. Integration of rational, trigonometric and irrational expressions	1	13	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8] - [11], additional [16] - [19]
14	Topic 14. Definite integral, properties, and computation. Improper Integrals	1	14	- Study and repetition of the material of lectures and practical classes	4	[2], [6], [8] - [11], additional [16] - [19]
15	Topic 15. FNP, limit continuity, partial derivatives and differentials	1	15	study and repetition of the material of lectures and practical classes	4	[2], [6], [8] - [11], additional [16] - [19]
16	Topic 16. Taylor's formula, FNP extreme. Quadratic matrix, Sylvester's criterion. Hessian and its application for the analysis of the curvature of a function	1	16	- Study and repetition of the material of lectures and practical classes	4	[2], [5], [6], [8] - [11], additional [16] - [19]
17	Topic 17. Method of least squares, conditional extremum,	1	17	- Study and repetition of the material of lectures and practical classes	4	[2], [5], [6], [8] - [11], additional [16] - [19]
18	Topic 18. Double Integral, Its Properties, and Computation	1	18	- Study and repetition of the material of lectures and practical classes - Preparation for the test	4	[2], [5], [6], [8] - [11], additional [16] - [19]
The total labor intensity of independent work in the discipline					90	-

4.4. Plan of extracurricular independent work, semester 2

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
Module 1. Probability Theory and Random Variable Theory						
1	Topic 1. Probability. Probability theorems	2	1	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
2	Topic 2. Discrete and continuous random variables	2	2	- Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
3	Topic 3. Numerical Characteristics of Discrete and Continuous Random Variables	2	3	- study and repetition of the material of lectures and practical classes; - Preparation for the test	4	[2], [11] - [13], additional [14], [15], [17]
4	Topic 4. Basic Laws of Random Variable Distribution	2	4	- Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
5	Topic 5. Random vectors	2	5	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
6	Topic 6. Numerical characteristics of random vectors. Correlation	2	6	Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
Module 3. Mathematical statistics and regression analysis						
7	Topic 7. Sampling. Estimating Sample Distributions	2	7	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
8	Topic 8. Point Estimates of Distribution Parameters	2	8	- Study and repetition of the material of lectures and practical classes - Preparation for the test	2	[2], [11] - [13], additional [14], [15], [17]
9	Topic 9. Interval estimates of distribution parameters	2	9	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
10	Topic 10. Parametric and nonparametric criteria	2	10	- Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
11	Topic 11. Regression analysis. Evaluating Statistical Dependencies	2	11	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
12	Topic 12. Bayesian regression	2	12	- Study and repetition of the material of lectures and practical classes - Preparation for the defense of an individual assignment	2	[2], [11] - [13], additional [14], [15], [17]

No p/n	Topic of the lesson	Semester	Completion Time (Weeks)	Type of independent work	Time Expenditure (Hours)	Educational and methodological support
13	Topic 13. Multivariate Probabilistic Models and Stochastic Processes	2	13	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
14	Topic 14. Markov Processes	2	14	- Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
15	Topic 15. Time series	2	15	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
16	Topic 16. Bayesian modeling	2	16	- Study and repetition of the material of lectures and practical classes	2	[2], [11] - [13], additional [14], [15], [17]
17	Topic 17. The Law of Large Numbers	2	17	- Study and repetition of the material of lectures and practical classes	4	[2], [11] - [13], additional [14], [15], [17]
18	Topic 18. Review: statistical methods of data analysis, experimentation	2	18	- study and repetition of the material of lectures and practical classes; - Preparation for the defense of an individual assignment	2	[2], [11] - [13], additional [14], [15], [17]
The total labor intensity of independent work in the discipline					54	-
Exam preparation					36	-
The total labor intensity of independent work in the discipline					90	-

V. EDUCATIONAL AND METHODOLOGICAL SUPPORT OF THE DISCIPLINE

5.1. References

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2. Klovo A.G., Lyapunova I.A. Mathematics in a technical university for the 2nd semester: a textbook / A.G. Klovo, I.A. Lyapunova; Southern Federal University. – Rostov-on-Don; Taganrog: Publishing House of the Southern Federal University, 2021. – 200 p. ISBN: 978-5-9275-3837-9. <https://hub.sfedu.ru/storage/1/1287888/3248ef6c-27ff-4d01-b21d-8bb81b0f7c02/>
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5.2. Further reading

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17. Collection of tasks for typical calculations and control works in mathematical disciplines [Text]. Part 1 / Editor-in-Chief A.I. Sukhinov; TRTU, EHF, Kaf. VM - Taganrog: TRTU Publ., 2006. - 545 p. (115 copies)
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1. Proceedings of Higher Educational Institutions. Mathematics (<https://kpfu.ru/science/nauchnye-izdaniya/ivrm>)
2. Mathematical Collection
(http://www.mathnet.ru/php/journal.phtml?jrnid=sm&option_lang=rus)
3. VMF. Series 1. Mathematics, mechanics
(http://www.mathnet.ru/php/journal.phtml?jrnid=vmumm&option_lang=rus)

5.4. List of Internet resources

1. <http://ictis-lms.sfedu.ru/>. E-learning system of ICTIB SFedU
2. <https://hub.sfedu.ru/>. SFedU Electronic Resources Portal

5.5. Scientific publications at A* level conferences, articles published in level 1 journals of the White List of scientific journals

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VI. MATERIAL AND TECHNICAL SUPPORT OF THE DISCIPLINE

The following premises, equipment and software are used in the implementation of the discipline:

No p/n	Type of classroom lesson	Recommended audience	
		Type	Equipment
1	Lectures	lecture-type auditorium	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office PowerPoint
2	Practical exercises	Seminar-type auditorium	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; – Microsoft Windows, Microsoft Office PowerPoint Microsoft Windows, Microsoft Office PowerPoint
3	Independent work	Audience of individual and group consultations, intermediate certification	– interactive whiteboard - 1 pc., teacher's computer - 1 pc.; Microsoft Windows, Microsoft Office PowerPoint Microsoft Windows, Microsoft Office PowerPoint

VII. METHODOICAL INSTRUCTIONS FOR STUDENTS ON MASTERING THE DISCIPLINE

7.1. General Provisions

It is recommended to familiarize yourself with the content of the course according to the work program of the discipline.

Write out (download) from the corresponding work program:

- list of recommended literature;
- the names of the lecture sections of the course;
- topics of practical classes;
- theoretical questions for the test and exam.

Students are recommended, in accordance with the schedule of lectures and practical classes in this discipline, to plan days of the week and hours for independent work, which will include preparation for lectures, practical classes, as well as preparation for intermediate control and exams.

7.2. Preparation for lectures (theoretical course)

Recommendations:

- before the next lecture, it is necessary to review the material of the previous lecture according to your notes;
- get acquainted with the content of the next lecture on the main sources of literature in accordance with the work program of the discipline;
- In case of difficulties in perceiving the material, it is necessary to turn to the main literary sources, lectures, to the lecturer - according to the schedule of his consultations, to the teacher - in practical classes.

7.3. Preparation for practical classes

Practical classes are aimed at consolidating theoretical knowledge, developing and forming practical skills and abilities in the course.

Recommendations:

- when preparing for a practical lesson, it is necessary to be guided by the content of the thematic plan in the work program of the discipline, i.e. to know the basic concepts, definitions, terminology, axioms and theorems;

- when preparing for the relevant practical lesson, it is necessary to look through the examples considered at the lecture from the lecture notes.

7.4. Preparation and implementation of the test

In accordance with the work program of the discipline "Higher Mathematics", the study of the relevant sections ends with the completion and submission of a test. The student should start performing the test work only after studying the theoretical material corresponding to the topic from the lecture notes or textbook, and analyzing the examples given. Options for tests are given in the fund of assessment tools (FOS) of the discipline "Higher Mathematics".

Recommendations for implementation:

- the student completes the version of the test provided by the teacher;
- the test is carried out on notebook sheets (up to 2-4 pages);
- when solving problems, it is necessary to justify each stage of the solution, to give the necessary formulas;
- calculations should be sufficiently detailed, arranged in a strict order;
- the solution of each problem should be brought to the answer required in the condition;
- the solution of problems of geometric content should be accompanied by a drawing or sketch, made accurately, indicating the axes of coordinates and the unit of measurement;
- Explanations for the problems should correspond to the designations given in the figures.

Tests must be done independently. Undone work deprives the student of the opportunity to check the degree of his preparedness on the topic under study. If the teacher determines that the work is not independent, namely, the student will not be able to answer the teacher's questions on solving the tasks of the test, then it will be evaluated for the minimum number of points.

Having received a checked work from the teacher, the student must independently work on mistakes. In case of non-credit for the work, the student can fulfill all the requirements as soon as possible and submit the work for re-checking, attaching the originally completed work. Retaking the test is carried out as part of retakes conducted by the teacher. In cases where a student encounters difficulties in studying theory or solving problems that he could not solve on his own, he has the opportunity to seek oral advice. Consultations are conducted by the teacher in accordance with the established schedule.

7.5. Independent work of students

Independent work of students involves preparation for practical, lectures, tests, the performance of recommended home/individual tasks in accordance with the schedule of independent work of the work program of the discipline "Higher Mathematics".

Recommendations:

- be guided by the schedule of independent work in the work program of the discipline;
- tasks must be completed in accordance with the number of the variant and drawn up in notebooks;
- to analyze unclear issues in classes and consultations;
- Preparation for the exam must be carried out on theoretical issues, the implementation and passing of tests.

7.6. Preparation for intermediate certification

Students who, for a good reason, could not score the required number of points in the current control of the module, can, in agreement with the teacher, get points before the start of intermediate certification by rewriting the tests. The main guideline in preparing for the exam is the questions given in the fund of assessment tools.

Intermediate certification is carried out on the basis of the results in accordance with the curriculum of the discipline.

It is possible to conduct intermediate certification in a remote form. In this case, all of the quizzes and exams that are listed in the discipline curriculum are completed using the Microsoft Teams platform. In this case, the exam is a rating control.

The topics of the questions and tasks of the exam in the discipline "Higher Mathematics" correspond to the mandatory minimum content on it. The exam ticket includes questions and tasks on sections and tasks considered during lectures, practical and laboratory classes. When preparing for the exam, it is recommended to use the sources specified in the educational and methodological support of the discipline.

The minimum number of points required for admission to the intermediate certification (exam) is 38 points (the maximum is 60 points). When summing up the results, the results of current and midterm control are taken into account, so it is important to attend classes regularly and complete all tasks on time.

A maximum of 40 points is given for the exam, a minimum of 22 points. The final score is made up of points for work during the semester, bonus points and points for the exam. The final score is converted into a grade as follows:

85-100 – "excellent", 71-84 – "good", 60-70 – "satisfactory", less than 60 – "unsatisfactory".

A student can receive up to 10 bonus points for successful participation in conferences, Olympiads, competitions in mathematics, as well as for completing additional tasks of increased complexity (including up to 5 bonus points for good and excellent results in the case championship in mathematics). Points are given by lecturers only if they receive admission to the exam (from 38 points).

VIII. FUND OF ASSESSMENT TOOLS

8.1. Passport of the Appraisal Tools Fund

Nop /n	Competency Achievement Indicator	Name of the Valuation Tool
1	CPC-1.1 - Solves standard professional problems using natural science and general engineering knowledge, methods of mathematical analysis and modeling	- Tests No1-3 in semesters 1-2 - Practical tasks in semester 2
2	CPC-1.2 - Conducts theoretical and experimental studies of objects of professional activity	- Exam questions and tickets
3	PC-1.2. MF-1.2. Applies the apparatus of probability theory, mathematical statistics and information theory to formulate and analyze artificial intelligence problems PC-1.3. MF-1.3 Applies the modern mathematical apparatus of probability theory to the study of machine learning methods and models.	- Tests No1-2 in semester 2 - Practical tasks in semester 2 - Exam questions and tickets
4	PC-2.1 MF-2.1 Uses Bayes' Theorem and its consequences, understands the differences between Bayesian statistics and the frequency approach	
5	PC-2.2 MF-2.2 Applies Bayesian estimation methods and Bayesian intervals to solve statistical problems and build models.	
6	PC-2.3 MF-2.4 - Applies Bayesian models to build and analyze regression, classification, clustering, improving the stability and interpretability of modern AI models	
7	SC 3.1 MF-3.1 Applies optimization techniques to the development and research of training algorithms	- Tests No1-3 in semesters 1-2 - Practical tasks in semester 2 - Exam questions and tickets
8	PC-4.1 MF-4.1 Applies statistical methods of analysis and machine learning to solve data analysis problems and conduct experiments on data	- Tests No1-2 in semester 2 - Practical tasks in semester 2 - Exam questions and tickets

8.2. Tests, semester 1

Tests are carried out in writing in practical classes. They can be implemented in the form of options with different levels of complexity at the discretion of the teacher.

Below are examples of multi-level tests.

Test No1. Version 1 - Simplified

1. The matrices A, B, C, D and numbers are given α and β . Find: a) AB , b) BD , c) $\alpha A^2 + \beta BC$, d) matrix rank D.

$$A = \begin{pmatrix} -5 & 1 & 2 \\ 3 & -1 & 4 \\ -1 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 3 & 4 & -2 & 6 \\ 0 & -1 & 1 & -3 \\ -2 & 5 & 2 & 1 \end{pmatrix}, C = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 5 & 1 \\ -4 & 4 & 7 \\ 1 & 6 & 5 \end{pmatrix}, D = \begin{pmatrix} -3 & 3 & 2 & 1 & 5 \\ 1 & 5 & 0 & 13 & 5 \\ 2 & 1 & -1 & 6 & 0 \\ -1 & 13 & 2 & -4 & 2 \end{pmatrix},$$

$$\alpha = -2, \beta = 3$$

2. Solve SLOUGH $\begin{cases} 3x - 5y = 17, \\ 4x + 3y = 5 \end{cases}$ methods a) Kramer, b) matrix, c) Gauss.

3. Solve SLOUGH $\begin{cases} 2x_1 + 3x_2 - x_3 = 3, \\ -x_1 + 2x_2 + x_3 = -1, \\ x_1 + 6x_2 + 3x_3 = 5. \end{cases}$

- 4) Points A(-5; 0;-1), B(2; 1;-2), C(-3; 1; 2), D(2; 1;-3). Find: a) cosine of angle B in triangle ABC, b) $\overrightarrow{AB} \cdot \overrightarrow{AC}$, c) $\overrightarrow{AB} \times \overrightarrow{AC}$, d) $\overrightarrow{AB} \cdot \overrightarrow{AC} \cdot \overrightarrow{AD}$, e) the area of the triangle ABC, f) the volume of the tetrahedron ABCD, equation g) a straight line AB, h) planes ABC.

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow the use of your math notes)	5 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	10 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	5 points
	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	
4.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	10 points
	Total Points	30 points

Test No1. Version 2

1. Matrices A, B, C, D and numbers are given α and β . Find: a) $\alpha A^2 + \beta BC$; b) the rank of the matrix D.

$$A = \begin{pmatrix} -5 & 1 & 2 \\ 3 & -1 & 4 \\ -1 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 3 & 4 & -2 & 6 \\ 0 & -1 & 1 & -3 \\ -2 & 5 & 2 & 1 \end{pmatrix}, C = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 5 & 1 \\ -4 & 4 & 7 \\ 1 & 6 & 5 \end{pmatrix}, D = \begin{pmatrix} -3 & 3 & 2 & 1 & 5 \\ 1 & 5 & 0 & 13 & 5 \\ 2 & 1 & -1 & 6 & 0 \\ -1 & 13 & 2 & -4 & 2 \end{pmatrix}, \alpha = -2, \beta = 3.$$

2. Solve the systems of equations: a) $\begin{cases} 2x_1 + 3x_2 - x_3 = 3, \\ -x_1 + 2x_2 + x_3 = -1, \\ x_1 + 6x_2 + 3x_3 = 5. \end{cases}$ b) $\begin{cases} 2x_1 + 4x_2 - 2x_3 + 6x_4 - 8x_5 = -4, \\ -x_1 - 2x_2 + x_3 - 3x_4 + 4x_5 = 2, \\ 5x_1 + 10x_2 - 2x_3 + 9x_4 - 8x_5 = -6. \end{cases}$

3. Points A(1; 2; 1), B(3; 0; 4), C(2; -2; 0), D(5; 4; 6) and $\alpha=1, \beta=2$. Find: a) the length of the segment AB; b) cosine of the angle B in triangle ABC; c) $\text{np}_{\overline{AB}}(\alpha \overline{BC} + \beta \overline{AD})$; d) $\overline{AB}$ and guide cosines of the

vector; $\overline{ABe}$ the area of the triangle ABC; (e) Height h_C of triangle ABC; g) the volume of the ABCD pyramid; h) the equation of the median in the triangle BCD lowered from the vertex B to the side CD.

4. Planes $(\Pi): Ax + By + Cz + D = 0$, $(\Pi'): A'x + B'y + C'z + D' = 0$, $(\Pi''): A''x + B''y + C''z + D'' = 0$ straight lines $(L): (x - x_0)/l = (y - y_0)/m = (z - z_0)/n$ are given; point $M(x'; y'; z')$: (a) construct the equation of the plane passing through the point M, and $\parallel$ -th plane (P); b) make an equation of a line passing through the point M, and $\parallel$ -th straight line (L); c) make an equation of the plane passing through the point M and the perpendicular line (L); d) make an equation of a straight line passing through the point M and a perpendicular line (P); e) make an equation of the plane passing through the point M and the line (L); f) find the point of intersection of a straight line (L) and a plane (P); g) make an equation of the plane passing through the point M and perpendicular to the planes (p') and (p''); h) find $\rho(M, (P))$. Here $(l; m; n) = (1; 5; -4)$:

(A; B; C; D)	(A'; B'; C'; D')	(A''; B''; C''; D'')	(x0; y0; z0)	(x'; y'; z')
(-5; 2; -1; 3)	(2; 1; -1; 4)	(-3; -1; 0; 1)	(2; -2; 3)	(2; -1; -3)

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow the use of your math notes)	5 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	10 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	5 points
4.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	10 points
	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	
	Total Points	30 points

Test No1. Version 3

Completing educational case tasks

1. Case: Linear transformations, eigenvalues, and vectors. A neural network with a linear layer is represented by a matrix

$$A = \begin{pmatrix} -5 & 1 & 2 \\ 3 & -1 & 4 \\ -1 & 0 & 1 \end{pmatrix}.$$

Define eigenvalues and eigenvectors.

Question: What is the geometric effect of such a layer on the input vectors? (Hint: Eigenvectors show directions that don't change direction after conversion.)

1.2.

2. Case: Systems of linear equations In the linear regression problem, it is necessary to select the weights w_1 , w_2 that satisfy the system:

$$\begin{cases} 2w_1 + w_2 = 4, \\ w_1 + 3w_2 = 5. \end{cases}$$

Task: solve the system analytically and find w_1 , w_2 . Explain how these ratios can be interpreted in the context of predicting the price of an object based on two criteria.

3. Case: Singular Decomposition and the Principal Component Matrix Method:

$$X = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 2 & 1 \end{pmatrix}.$$

Problem: Find the XTX covariance matrix, calculate the eigenvalues, and indicate the direction of the principal component.

The question is: how will the orientation of the data change if only the first component is left?

4. Case: There are 4 programmers working in a startup for the development of a cloud platform: Alexey, Boris, Victor and Grigory. Each of them offers their own approach to four key aspects of development:

- system architecture,
- error handling,
- scalability,
- security.

Estimates of the impact of each approach are presented in a **compatibility matrix**, where each row is an assessment of a single programmer's approach for each aspect.

To determine whether the **team could develop an independent and sustainable architecture**, the project architect decided to calculate **the determinant of this matrix**.

$$\begin{vmatrix} 3 & -1 & 2 & 0 \\ 1 & -2 & 4 & 3 \\ 5 & 3 & 6 & 2 \\ 4 & 1 & 1 & -3 \end{vmatrix}$$

1. Calculate the determinant of this matrix:

- a) by means of decomposition by row or column;
- b) by reducing to a triangular form (for example, by the Gaussian method).

2. Draw a conclusion:

- If the determinant $\neq 0$, all approaches are independent and the team is flexible.
- If the determinant $= 0$, there is a linear dependence in the approaches, and a revision of the methods is needed.

5. Case: A small startup team has three programmers: Alina, Boris, and Konstantin. They distribute three types of tasks among themselves:

- writing code,
- testing,
- code review.

The project manager has compiled a system of equations that reflects how many hours each employee spends on each category of tasks, as well as the total workload in hours, where:

- x_1 — Alina's contribution,
 - x_2 — Boris's contribution,
 - x_3 is Konstantin's contribution.
1. Determine the individual contribution of each programmer:
 - a) according to **Kramer's rule**;
 - b) **by the Gaussian method** (reduction to a stepped form);
 - c) **by the matrix method** (through an inverse matrix).
 2. Draw a conclusion:
 - Who in the team is the most busy?
 - Is it possible to redistribute tasks so that the load is even?

$$\begin{cases} 2x_1 + 3x_2 - x_3 = 3, \\ -x_1 + 2x_2 + x_3 = -1, \\ x_1 + 6x_2 + 3x_3 = 5. \end{cases}$$

6. Case: A large IT company has a team of 4 programmers, each of whom writes modules that interact with each other. To assess the performance of the system, the architects built several matrices:

- Matrix A reflects how efficiently each programmer solves three types of problems: algorithmic, interface, and architectural.
- Matrix B (3×4) shows how much each task type affects the four key components of the project.
- The C matrix describes the impact of each component of the project on the various stages of development (design, development, testing).
- Matrix D is a matrix of dependencies between different components of a project. The higher the interdependence, the more difficult it is to scale the project.

The team of analysts wants to calculate a combined estimate of the complexity of the project using the formula: $M = \alpha \cdot A^2 + \beta \cdot (B \cdot C)$, where

- A^2 is the interaction between programmers based on their task types,
- $B \cdot C$ is the impact of tasks on development stages,
- $\alpha = -2$, $\beta = 3$ are the weights of importance of internal and external complexity.

You also need to determine the rank of the matrix D to understand how much the components depend on each other (rank = number of independent components).

1. Calculate the matrix M, which represents the total complexity of the project.
2. Determine the rank of the matrix D.
3. Conclude: if the rank is less than 5, then some components of the project can be combined – what benefits will this give in development management?

$$A = \begin{pmatrix} -5 & 1 & 2 \\ 3 & -1 & 4 \\ -1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 & -2 & 6 \\ 0 & -1 & 1 & -3 \\ -2 & 5 & 2 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & -1 & 3 \\ 2 & 5 & 1 \\ -4 & 4 & 7 \\ 1 & 6 & 5 \end{pmatrix},$$

$$D = \begin{pmatrix} -3 & 3 & 2 & 1 & 5 \\ 1 & 5 & 0 & 13 & 5 \\ 2 & 1 & -1 & 6 & 0 \\ -1 & 13 & 2 & -4 & 2 \end{pmatrix}, \quad \alpha = -2, \quad \beta = 3;$$

No back.	Assessment criteria	Maximum Points
1.	Written work includes 6 cases, can be done both individually and in teams of 2-4 people Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; performance of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors; performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
2.		5 points
3.		5 points
4.		5 points
5.		5 points
6.		5 points
	Total Points	30 points

Test No2

Version 1 - Simplified

1. Calculate:

$$1) \lim_{n \rightarrow \infty} \frac{\sqrt{n^3 - 1} + 2n}{(\sqrt{n} + 1)^3}; \quad 2) \lim_{x \rightarrow \infty} (\sqrt{x+1} - \sqrt{x}); \quad 3) \lim_{x \rightarrow 0} \frac{\arcsin 5x}{\operatorname{tg} 3x}.$$

2. Will the points $x=-2$, $x=0$ be the points of the discontinuity of the function (prove):

$$f(x) = \begin{cases} x+4, & x \leq -2 \\ 1+(x+1)^2, & -2 < x < 0 \\ -x, & x \geq 0 \end{cases}$$

3. Points $x=0$, $x=2$ are for the $y = \frac{4}{x(x-2)}$ 1) Points of discontinuity of the second kind.

2) Jump points. 3) Points of removable gap. 4) Points of continuity.

4. 1) If $y = \sin 3x - 3 \cos 2x - \frac{1}{x}$, then $y' = ?$

2) If $y = 2^x \cdot 3^{-x} + \ln x + \operatorname{tg} 3x$, then $y' = ?$

3) If $\sqrt{x} + \sqrt{y} + xy = 2$, then $y'_x = ?$

4) If $\begin{cases} x = e^{2t}, \\ y = \arcsin 3t, \end{cases}$ then $y'_x = ?$

5. If $f(x) = \sqrt{x}$, then the equation of the tangent to the graph of the function $f(x)$ at $M\left(\frac{1}{4}, \frac{1}{2}\right)$ has the form _____.

6. According to L'Hôpital's rule $\lim_{x \rightarrow 1} x^{\frac{1}{1-x}}$, the following is equal _____

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow you to use your math note)	5 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	6 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	6 points
4.		6 points
5.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	6 points
6.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	6 points
	Total Points	35 points

Test No2 Version 2

1. Calculate:

$$1) \lim_{x \rightarrow x_0} \frac{-5x^2 - 6x - 1}{2x^2 + 3x + 1}, a) x_0 = -\frac{1}{5}; b) x_0 = -\frac{1}{2}; c) x_0 = -1; d) x_0 = \infty;$$

$$2) \lim_{x \rightarrow \infty} \left(\frac{5x+2}{5x-2} \right)^{2x+5}; 3) \lim_{x \rightarrow 3} \frac{\sqrt{2x-1} - \sqrt{5}}{x-3}; 4) \lim_{x \rightarrow 0} \frac{tg^2 \frac{x}{2}}{x^2}.$$

$$2. \text{ To investigate for continuity: } 1) f(x) = \begin{cases} -x, & x \leq 0 \\ 1 - (x-1)^2, & 0 < x \leq 2 \\ 2, & x > 2 \end{cases} : 2) f(x) = 6^{\frac{x}{x+5}}.$$

$$3. e^{x-y} + xy = 0, y'_x - ?$$

$$4. y = \arcsin 2x \cdot (x-3)^{(x-3)}, y' - ?$$

$$5. \begin{cases} x = \arcsin t \\ y = \ln(1-t^2) \end{cases}; y''_{xx} - ?$$

$$6. \text{ Calculate according to L'Hôpital's rule: } \lim_{x \rightarrow 0} (\sin x)^{\arctg x}; \lim_{x \rightarrow 1} (x-1) \ln \sin \pi x.$$

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow you to use your math note)	5 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	6 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	6 points
4.		6 points
5.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	6 points
6.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	6 points
	Total Points	35 points

Test No2. Version 3

Completing educational case tasks

1. Case: Numeric sets, faces, and feature normalization

The "age" attribute takes values from 18 to 65. It is necessary to bring it to the range [0, 1].

Task:

Write down the normalization formula and calculate the result for age 40.

(min = 18, max = 65)

Question:

How will the result change if the range is increased to [0, 100]?

2. Case: Sequences and Their Limit

The model is trained in gradient descent increments:

$w_{n+1} = 0.9w_n + 0.1$.

Task

:Prove that the sequence converges and find its limit.

Question:

What does this limit mean in the context of stabilizing the weights of a neural network?

2. Case: Infinitesimal quantities

During training, the θ parameter changes to $\Delta\theta = 0.0001$. The error changes to $\Delta L = 0.0003$.

Problem: Show that the relation $\Delta L / \Delta\theta$ tends to the derivative $dL/d\theta$ when $\Delta\theta \rightarrow 0$.

The question is: how to interpret this as the local sensitivity of the loss function?

4. Case: Newton's binomial and the number e

To calculate the exponent, the activation function uses an approximation:

$e^x \approx 1 + x + x^2/2$.

Task: for $x = 0.1$, calculate the approximation value and compare it with the real $e^{0.1}$.

Q: Why does using Newton's binomial help optimize computations in embedded models?

5. Case: Increment and extremes

$L(x) = x^3 - 6x^2 + 9x$.

Task: find the intervals of increasing and decreasing.

Question: How does this help to find the decreasing areas of the loss function?

6. Case: Research and Charting

$L(x) = x^3 - 6x^2 + 9x + 1$.

Task: Investigate the function (increase, extremes, convexity, asymptotes).

Q: How does the form of the loss function graph help you understand where learning gets stuck in local lows?

No back.	Assessment criteria	Maximum Points
1.	Written work includes 6 cases, can be done both individually and in teams of 2-4 people Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; performance of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors; performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
2.		6 points
3.		6 points
4.		6 points
5.		6 points
6.		6 points
	Total Points	35 points

Test No3

Version 1 - Simplified

- Find the integrals: $\int \left(\frac{5}{\cos^2 x} - \frac{7}{\sqrt[3]{x^5}} + \frac{3}{\sqrt{x^2+49}} + 8 \sin x - \frac{3}{x} \right) dx$.
- $\int \frac{e^x dx}{\sin^2(e^x+2)}$.
- $\int (x^2 - 5x - 2) \sin 2x dx$.
- $\int \frac{(6x-1)}{x^2-6x+13} dx$.
- Find the integrals: a) $\int \sin^3 x \cos^3 x dx$, b) $\int \sin 5x \sin x dx$; c) $\int \frac{dx}{2-2\sin x - \cos x}$.

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow you to use your math note)	7 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	7 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	7 points
4.		7 points
5.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	7 points
	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	
	Total Points	35 points

Test No3 Version 2

- Find the integral: $\int \left(3^x + \frac{9}{\sqrt{81-x^2}} + \frac{2}{\sqrt{x^2-16}} - 6 \sin x \right) dx$.
- Find the integrals: a) $\int \frac{dx}{\cos^2(3x-10)}$, b) $\int \frac{3^x}{\sqrt{5+2 \cdot 3^x}} dx$
- Find the integrals: a) $\int x^7 \ln x dx$, b) $\int (x^2 + 6x - 3) \cos 4x dx$, c) $\int e^{3x} \sin 2x dx$.
- *. Find the integrals: a) $\int \frac{(x-10)}{x^2-8x+20} dx$, b) $\int \frac{(6x-1)dx}{x^2+2x-15}$; c) $\int \frac{(2x-9)dx}{(x-2)(x^2+x+4)}$.
- Find the integrals: a) $\int \sin^5 x \cos^2 x dx$, b) $\int \sin 3x \sin 7x dx$; c) $\int \frac{dx}{1-\sin x-2 \cos x}$.
- Find the integrals: a) $\int x^2 \sqrt{16-x^2} dx$, b) $\int \frac{(x-3)dx}{\sqrt{x^2+4x+8}}$.

No Rear	Assessment criteria	Maximum Points
1.	Written work done in the classroom for 90 minutes (you can allow you to use your math note)	5 points
2.	Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings;	6 points
3.	Completion of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	6 points
4.	Completion of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors;	6 points
5.		6 points
6.	Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	6 points
	Total Points	35 points

Test No3. Version 3
Completing educational case tasks

Case 1. Primetypive, indefinite integral

Situation: When training a model, you need to evaluate the accumulated error over time.

Task:

Let the instantaneous error of the model in time t be described by the function $E'(t) = 2 \cdot t + 3$. Find the total error $E(t)$.

Case 2. Replacing a variable when calculating the expected value of an error function

Situation: When estimating the data distribution loss function, a change from one variable to another is required.

Task:

Calculate integral $\int x / (x^2 + 1) dx$.

Case 3. Calculating the average model error on an interval

Situation: The model error function at epoch t is described by $L(t) = t^2$.

Task:

Find the average error during training $t \in [0, 2]$: $\bar{L} = 1/(2-0) * \int_0^2 t^2 dt$.

Case 4. Analysis of the convergence of the loss function by two parameters

Situation: The model error depends on two parameters x and y : $E(x, y) = x^2 + x \cdot y + y^2$.

Task:

Find partial derivatives E_x and E_y , determine the direction of the greatest error reduction.

Case 5. Finding the Optimum of the Loss Function in Hessian

Situation: The model's loss function is $L(x, y) = x^2 + 4xy + 5y^2$.

Task:

Check whether point $(0, 0)$ is the minimum of the function using the second differential matrix.

Case 6. Two-Feature Optimization and Error Area Estimation

Situation: The model is trained on two traits, x and y , and the error is expressed as $E(x, y) = x^2 + y^2$.

Task:

Calculate the cumulative error in the square area $x, y \in [0, 1]$:

$I = \int_0^1 \int_0^1 (x^2 + y^2) dx dy$.

No back.	Assessment criteria	Maximum Points
1.	Written work includes 6 cases, can be done both individually and in teams of 2-4 people Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; performance of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors; performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	5 points
2.		6 points
3.		6 points
4.		6 points
5.		6 points
6.		6 points
	Total Points	35 points

Test No3 Version 4 (in the form of a colloquium)

Fam _____ Name _____ Father _____

Group _____ Signature _____

Enter the answers as a sequence of numbers

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

1) Let $A = \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 3 \\ -2 & 5 \end{pmatrix}$. Indicate in ascending order the numbers of the correct equations.

$$1. \quad A + 2B = \begin{pmatrix} 1 & 4 \\ 0 & 11 \end{pmatrix} \quad 2. \quad 2A - 3B = \begin{pmatrix} 5 & -15 \\ 14 & -13 \end{pmatrix} \quad 3. \quad AB = \begin{pmatrix} 1 & -1 \\ -6 & 17 \end{pmatrix}$$

$$4. \quad BA = \begin{pmatrix} 9 & -1 \\ 14 & 9 \end{pmatrix} \quad 5. \quad A^2 = \begin{pmatrix} 1 & -8 \\ 8 & -7 \end{pmatrix} \quad 6. \quad B^2 = \begin{pmatrix} -5 & 12 \\ -8 & 19 \end{pmatrix}$$

2) Indicate the numbers (in ascending order) of determiners equal to (-1) .

$$1. \quad \begin{vmatrix} 1 & -3 \\ -3 & 10 \end{vmatrix} \quad 2. \quad \begin{vmatrix} 6 & -5 \\ -5 & 4 \end{vmatrix} \quad 3. \quad \begin{vmatrix} 10 & 3 \\ -3 & -1 \end{vmatrix}$$

$$4. \quad \begin{vmatrix} -1 & -3 & 0 \\ -3 & -7 & 2 \\ 0 & 1 & 3 \end{vmatrix} \quad 5. \quad \begin{vmatrix} -2 & 1 & -2 \\ -1 & 5 & -4 \\ 3 & -1 & 0 \end{vmatrix} \quad 6. \quad \begin{vmatrix} 0 & -1 & 0 \\ 7 & 0 & -5 \\ -3 & -2 & 2 \end{vmatrix}$$

3) Indicate the numbers (in ascending order) of the matrices for which the inverses exist.

$$1. \begin{pmatrix} 4 & 8 \\ 3 & 6 \end{pmatrix} \quad 2. \begin{pmatrix} 4 & -1 \\ 8 & 2 \end{pmatrix} \quad 3. \begin{pmatrix} 8 & 5 \\ 5 & 3 \end{pmatrix} \quad 4. \begin{pmatrix} 6 & 3 \\ -4 & 2 \end{pmatrix} \quad 5. \begin{pmatrix} 3 & 12 \\ 1 & 4 \end{pmatrix} \quad 6. \begin{pmatrix} 3 & 12 \\ 1 & 4 \end{pmatrix}$$

4) Specify the numbers (in ascending order) of the matrix pairs that can be multiplied in the order listed.

$$1. \begin{pmatrix} 4 & 3 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix} \quad 2. \begin{pmatrix} 0 & 3 & 5 \\ 9 & 12 & -2 \end{pmatrix} \begin{pmatrix} 3 & 4 & 0 \\ 9 & 0 & -4 \\ -3 & 2 & 18 \end{pmatrix} \quad 3. \begin{pmatrix} 14 & 2 \\ 7 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \\ 8 & 2 & -1 \end{pmatrix}$$

$$4. \begin{pmatrix} 5 & 7 & 4 \\ 5 & 4 & 4 \end{pmatrix} \begin{pmatrix} -3 & 6 \\ 12 & 9 \end{pmatrix} \quad 5. \begin{pmatrix} 1 \\ 6 \end{pmatrix} \begin{pmatrix} 1 & 8 & 0 \\ 7 & 1 & -4 \end{pmatrix} \quad 6. \begin{pmatrix} 1 & 8 & 5 \\ 5 & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 7 & -3 \end{pmatrix}$$

5) Indicate the numbers (in ascending order) of the incompatible SLUAs.

$$1. \begin{cases} 3x - 9y = 15, \\ x - 3y = 5. \end{cases} \quad 2. \begin{cases} x + 6y = 8, \\ 3x + 18y = 23. \end{cases} \quad 3. \begin{cases} 5x + 9y = 12, \\ 9x - 3y = 17. \end{cases}$$

$$4. \begin{cases} 7x = 2, \\ 4x - 5y = 3, \\ 3x - 5y + 7z = 2. \end{cases} \quad 5. \begin{cases} 2x - 5y + 7z = 4, \\ 4x - 8y + 3z = -1, \\ 5x - 3y + 2z = 4. \end{cases} \quad 6. \begin{cases} 2x - y + 4z = 2, \\ 6x - 3y + 12z = 5, \\ 7x - 2y + 4z = 3. \end{cases}$$

6) Indicate the numbers (in ascending order) of the SLOUGH that can be solved by the Kramer method.

$$1. \begin{cases} 2x - y + 2z = 3, \\ 7x + 3y - 4z = 5. \end{cases} \quad 2. \begin{cases} 5x - y = 1, \\ 5x + y = 3. \end{cases} \quad 3. \begin{cases} 3x - 2y = 1, \\ 6x - 4y = 2. \end{cases}$$

$$4. \begin{cases} x - 5y + 2z = 3, \\ 3x + 3z = 2, \\ 4x + 7z = 9. \end{cases} \quad 5. \begin{cases} 2x - 3y + 4z = -2, \\ 6x - 4y + 2z = 7, \\ 8x - 7y + 6z = -3. \end{cases} \quad 6. \begin{cases} 8x - y - 5z = -1, \\ -2x - 4y + 3z = 2, \\ 7x - 5y - 2z = -3. \end{cases}$$

7) Indicate in ascending order the numbers of correct statements.

- 1) Non-collinear vectors $\vec{a}$ and $\vec{b}$ their vector product $\vec{c}$ form the right triple of vectors.
- 2) The modulus of the sum of vectors is always less than the sum of their modulus.
- 3) The angle between two vectors is obtuse if and only if their dot product is negative.
- 4) If you swap 2 vectors, their dot product will change sign.
- 5) If you swap 2 vectors, their vector product will change sign.
- 6) The mixed product of the three non-coplanar vectors is equal to the volume of the box constructed on these vectors.

8) Indicate in ascending order the numbers of operations that result in the vector.

$$1. (\vec{a} \times \vec{b}) \cdot \vec{c} \quad 2. (5\vec{a}) \times \vec{b} \quad 3. (\vec{a} - 7\vec{b}) \times \vec{c} \quad 4. (\vec{a}, \vec{b}, \vec{c} + 5\vec{b}) \quad 5. (\vec{a} \times \vec{b}) \times (2\vec{c}) \quad 6. (\vec{a} - 2\vec{b}) \cdot \vec{c}$$

9) Indicate in ascending order the numbers of the equations of the line on the plane.

$$1. 2x^2 + y = 3 \quad 2. 2x - \frac{y}{9} = 15 \quad 3. 0 \cdot x + 0 \cdot y = 7 \quad 4. 3x + 2y = 7 \quad 5. xy = 5 \quad 6. \frac{x}{y} = 1$$

10) Indicate in ascending order the numbers of pairs of parallel lines on the plane.

1. $2x - y = 1$ and $x - 2y + 5 = 0$

2. $x + 2y = 9$ and $-2y - x - 3 = 0$

3. $2x - y - 6 = 0$ and $\frac{x+2}{-1} = \frac{y}{2}$

4. $y - 2x = -5$ and $\begin{cases} x = t + 3, \\ y = 2t, \end{cases} \quad -\infty < t < +\infty$

5. $6x - 4y - 5 = 0$ and $\frac{x+1}{2} = \frac{y+2}{3}$

6. $2x + y + 2 = 0$ and $\begin{cases} x = t + 2, \\ y = 2t - 1, \end{cases} \quad -\infty < t < +\infty$

11) Indicate in ascending order the numbers of the equations of the plane in space.

1. $0 \cdot x + 0 \cdot y - 2z = 3$

2. $x + 2y + z^2 = 2$

3. $0 \cdot x = 5$

4. $x + 6y + 4z = 10$

5. $3x^2 - z = 0$

6. $2x - 3y + 7z = 6$

12) Specify the vectors $\vec{a}\{14;8;-42\}$, $\vec{b}\{4;6;-2\}$, $\vec{c}\{-5;3;1\}$, $\vec{d}\{-1;3;4\}$, and planes:

$\pi_1: 4x - 3y + 2z - 3 = 0$, $\pi_2: 3x - 2y - 7 = 0$, $\pi_3: 5x - 3y + z - 7 = 0$, $\pi_4: x + y + 2z - 3 = 0$. Specify the numbers of the correct statements in ascending order.

1) $\vec{a} \perp \pi_1$ 2) $\vec{b} \perp \pi_2$ 3) $\vec{c} \perp \pi_3$ 4) $\vec{d} \perp \pi_4$ 5) $\vec{a} \perp \pi_2$ 6) $\vec{c} \perp \pi_4$

13) Indicate in ascending order the numbers of converging numerical sequences.

1. $x_n = 7 - (-1)^{n-1}$, $n \in \mathbb{N}$

2. $x_n = \frac{\cos n}{3-n}$, $n \in \mathbb{N}$

3. $x_n = \frac{9^2}{n^2}$, $n \in \mathbb{N}$

4. $x_n = 2^{n-5}$, $n \in \mathbb{N}$

5. $x_n = \frac{5}{n^{1,2}}$, $n \in \mathbb{N}$

6. $x_n = (-1)^{n+1}$, $n \in \mathbb{N}$

14) Indicate the numbers (in ascending order) of the correct formulas.

1) The function $f(x) = \frac{1-7x^2}{5-2x}$ has an asymptote $x = 5$.

2) The function $f(x) = \frac{x^2-5x-7}{x-3}$ has an asymptote $y = x - 2$.

3) The function $f(x) = \frac{2x^4+x^2}{|x|-7}$ has an asymptote $x = -7$.

4) The function $f(x) = (3x+4) \cdot e^{2x-5} - 9$ has an asymptote $y = -9$.

5) The function $f(x) = \frac{x^3-5x^2-6}{x-1}$ has an asymptote $y = x - 5$.

6) The function $f(x) = \frac{(2x-3) \cdot \sin x}{x^2-3x+7} + 5$ has an asymptote $y = -5$.

15) Indicate the numbers (in ascending order) of the correct formulas.

1. $(x^5)' = 5x^5$

2. $(\sin 2x)' = 2 \cos 2x$

3. $(\arccos x)' = \frac{1}{\sqrt{1-x^2}}$

4. $(\lg x)' = \frac{1}{x \cdot \lg 10}$

5. $(\pi)' = 0$

6. $(\operatorname{tg} x)' = \operatorname{ctg} x$

16) Indicate the numbers (in ascending order) of the correct formulas.

1. $d(\sin 2x) = -2 \cos 2x dx$ 2. $d(-2x^3) = -6x^2 dx$ 3. $d(\ln(2x)) = \frac{dx}{x}$

4. $d^2(\ln x^3) = \frac{3}{x^2} dx^2$

5. $d^2(4x^4) = 48x^3 dx^2$

6. $d^2(\sin 2x) = 2 \sin 2x dx^2$

17) Indicate the numbers (in ascending order) of the correct transformations.

$$1. \lim_{x \rightarrow \infty} \frac{1-x^2}{\ln 3x} = \lim_{x \rightarrow \infty} \frac{(1-x^2)'}{(\ln 3x)'}$$

$$2. \lim_{x \rightarrow +0} \frac{x^2-1}{\cos x} = \lim_{x \rightarrow +0} \frac{(x^2-1)'}{(\cos x)'}$$

$$3. \lim_{x \rightarrow +0} \frac{1-x^2}{\ln 3x} = \lim_{x \rightarrow +0} \frac{(1-x^2)'}{(\ln 3x)'}$$

$$4. \lim_{x \rightarrow \pi} \frac{1+\cos 3x}{2\sin x} = \lim_{x \rightarrow \pi} \frac{(1+\cos 3x)'}{(2\sin x)'}$$

$$5. \lim_{x \rightarrow +\infty} \frac{x^{-1}}{\ln x} = \lim_{x \rightarrow +\infty} \frac{(x^{-1})'}{(\ln x)'}$$

$$6. \lim_{x \rightarrow \pi} \frac{1-\cos 3x}{2\sin x} = \lim_{x \rightarrow \pi} \frac{(1-\cos 3x)'}{(2\sin x)'}$$

18) Indicate, in ascending order, the numbers of functions that are primitive for function $y = 4 \sin(2x)$.

$$1. F(x) = 2(1 - \cos(2x))$$

$$2. F(x) = 4 \sin^2 x$$

$$3. F(x) = 7 - \cos(2x)$$

$$4. F(x) = 5 + 2 \sin(2x)$$

$$5. F(x) = 9 - 4 \cos^2 x$$

$$6. F(x) = 9 + 4 \cos^2 x$$

19) Indicate in ascending order the numbers of indefinite integrals from rational functions.

$$1. \int \frac{2x^2 - x - 3}{x+2} dx$$

$$2. \int \frac{x-5}{3-\cos x} dx$$

$$3. \int \frac{x^3}{x-1} dx$$

$$4. \int \frac{x^2-1}{x+1} dx$$

$$5. \int e^{\frac{-6}{x+1}} dx$$

$$6. \int \frac{x^2-x}{x+5} dx$$

20) Indicate in ascending order the numbers of correct statements.

1) A definite integral $\int_2^4 \frac{x^2-16}{x+1} dx$ is equal to a positive number.

2) A definite integral $\int_0^\pi \sin x dx$ is equal to a positive number.

3) A definite integral $\int_0^\pi \cos x dx$ is equal to a positive number.

4) A definite integral $\int_1^2 \frac{x^2-1}{x+3} dx$ is equal to a positive number.

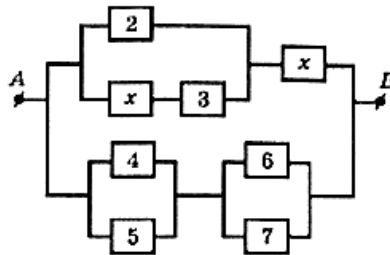
Assessment criteria: tasks 1-5 are evaluated by 1 point, tasks 6-20 are evaluated by 2 points.

8.3. Semester 2 tests

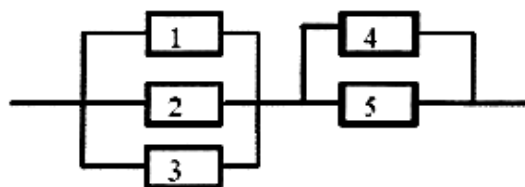
Test No1

The test is carried out in the form of an online test in LMS Moodle.

1. In Carlson's fridge there are 2 jars of raspberry jam, 10 jars of cherry jam and 5 jars of apricot jam. Karlsson chooses 3 jars at random to eat for breakfast.
2. The number in the cell indicates the number of ways in which it can be walked. How many ways can it be passed from A to B if $x = 10$?



3. There are 10 white, 15 black, 12 red and 8 blue balls in the urn. 4 balls are taken out at random. What is the probability that 2 white balls and at least 1 blue one are taken out. The answer should be rounded up to thousandths.
4. A rhombus and a circle are inscribed in a square with a side of 5 cm, and the vertices of the rhombus lie in the middle of the sides of the square. A dot is thrown into the square at random. What is the probability that she will hit the circle but not hit the diamond? The answer should be rounded up to thousandths.
5. Defects in the production of the product due to defect A are 10%, due to defect B - 8%. The plant produces 88% of suitable products. Find the probability that a randomly selected product will have defect A if it has defect B.
6. Two automatic machines produce identical parts, which go to a common conveyor. The probability of obtaining a non-standard part on the first machine is 0.075, and on the second - 0.09. The productivity of the second machine is twice as high as the first. Find the probability that a random part taken from the assembly line is non-standard.
7. The diagram of connecting the elements forming a circuit with one input and one output is considered. Element failures are assumed to be independent events in the aggregate. The failure of any of the elements leads to an interruption of the signal in the branch of the circuit where this element is located. The failure probabilities of elements 1, 2, 3, 4, 5, 6, respectively, are 0.4; 0.2; 0.5; 0.4; 0.4; 0.2. Find the probability that the signal will pass from entry to exit. The answer is rounded up to the nearest hundredth.



8. A worker maintains 10 machines of the same type. The probability that the machine will require the attention of the worker within an hour is 0.05. Find the probability that there will be three to five of these requirements within an hour. The answer should be rounded up to thousandths.

9. The probability that a bad but diligent basketball player will throw the ball into the basket is 0.3. 150 shots were made. Find the probability that there will be at least 50 hits. The answer should be rounded up to thousandths. (Solve the problem without using Bernoulli's formula.)

10. Specify the conditions under which the Poisson formula can be applied to rare events:

a) n little; b) n is great; c) $NPQ < 9$; d) $5 < NPQ < 20$; e) $p < 0.1$; (e) $p > 0.5$; g) $np < 500$.

No back.	Assessment criteria	Maximum Points
1.	The test is conducted online within 90 minutes	1 point
2.		1 point
3.	The results are evaluated automatically by the LMS	1 point
4.		1 point
5.	The criterion for the correctness of the answer is its coincidence with the reference one	1 point
6.		1 point
7.		1 point
8.		1 point
9.		1 point
10.		1 point
	Total Points	10 points

Test No2

The test is carried out in the form of an online test in LMS Moodle.

1. In the money lottery, 100000 tickets have been issued, which are distributed free of charge. 10 prizes of 100 thousand rubles each, 50 prizes of 10 thousand rubles, 1000 prizes of 500 rubles and 10000 prizes of 100 rubles are drawn. In the answer, indicate the mathematical expectation of the amount of winnings (in rubles).

2. Independent discrete random variables X and Y are given by the laws of distribution.

X	-10	-5	1
P	p	0,61	0,11

Find the variance of the random variable $Z = 10X + 3Y$.

3. The device consists of 1000 cells that work independently of each other. The probability of failure of any element during time T is 0.005. To construct a series of distributions of the number of X elements that have failed during time T . Find the mathematical expectation $M(X)$ and the standard deviation $\sigma(X)$ of this random variable. In the answer, indicate the value of $M(X) + 3\sigma(X)$, rounded to the nearest higher integer value.

4. The distribution law of the discrete random variable **X** is as follows:

The
is $M(X)$

Y	-15	10
P	0,23	q

values of x and y are natural numbers, with $x < y$. The expected value of **X** $= -1.1$, and the variance of $D(X) = 19.89$. Find y .

5. The

continuous random variable **X** is given by the distribution function:

$$F(x) = \begin{cases} 0 & , \quad x < -5 \\ a\sqrt{x+6} + b & , \quad -5 \leq x \leq 10 \\ 1 & , \quad x > 10 \end{cases}$$

Find the unknown coefficients a , b , and probability P that the random variable **X** will fall within the range $(-3; 5)$. Round the result to the nearest thousandth.

6. The continuous random variable **X** is given the probability density:

$$f(x) = \begin{cases} 0 & , \quad x < -2 \\ 1/8 & , \quad -2 < x < 0 \\ C(1-x/4) & , \quad 0 < x < 4 \\ 0 & , \quad x > 4 \end{cases}$$

It is known that the constant $C > 0$. Find the unknown coefficient C and the variance $D(X)$. In response, write down $D(X)$.

7. The random variable X has a normal distribution law ($MX = 100$; $DX = 250$). Find the probability of an event $\{X \in (90, 120)\}$. Round the result to nearest hundredth.

X / Y	c	d
a	0,2	0,1
b		0,3

the

8. Random vector **(X; Y)** is uniformly distributed in a rectangular region and a probability density

has

$$f(x, y) = \begin{cases} S & , \quad x \in [a, b], y \in [c, d], \\ 0 & , \quad \text{в противном случае.} \end{cases}$$

X	-10	x	y
P	0,2	0,7	p

It is known that $a = -10$, $b = 20$, $c = -5$, $d = 15$. Find the probability of an event $\{-5 \leq X \leq 7; -1 \leq Y \leq 7\}$. Round the answer to the nearest hundredth.

9. The system of two discrete random variables (X, Y) is given by the distribution table:

In this case, $a = -2$, $b = 4$, $c = -1$, $d = 5$. Find the correlation coefficient of random variables **X** and **Y**. Round the result to the nearest hundredth.

10. The random variables X , Y , and Z are independent in pairs. Then the statement " X , Y , and Z are independent in the aggregate"

- a) it is always fair; b) it is not always fair;
c) unfairly; d) fair in some cases.

No back.	Assessment criteria	Maximum Points
1.	The test is conducted online within 90 minutes	2 points
2.		1 point
3.	The results are evaluated automatically by the LMS	1 point
4.		2 points
5.	The criterion for the correctness of the answer is its coincidence with the reference one	1 point
6.		2 points
7.		1 point
8.		1 point
9.		2 points
10.		2 points
	Total Points	15 points

8.4. Individual assignments, semester 2

Individual task No1

An individual task is performed by the student in writing (it is possible to submit it in the form of an Excel file or a Python program) and handed over to the teacher for checking, after which, if necessary, a conversation is held with the student to solve problems.

Task 1. Build a sample containing 200 elements (file Selection_1.xlsx). The resulting selection requires:

- 1) build a grouped statistical series;
- 2) build a histogram and a polygon of frequencies (relative frequencies);
- 3) to construct an empirical distribution function;
- 4) find estimates of distribution parameters (sample mean, corrected variance and deviation, mode, median);
- 5) construct confidence intervals for estimates of mathematical expectation and variance of the general population;
- 6) on the basis of the preceding points, to put forward a hypothesis about the type of the law of distribution of the general population;
- 7) using the Pearson criterion of agreement, test the hypothesis from point 6.

Task 2. Build a sample of 10 values of feature pairs (file Selection_2.xlsx). The resulting sample requires:

- 1) build a correlation field;
- 2) to estimate the coefficients of paired linear regression;
- 3) to check the level of significance of linear regression as a whole;
- 4) build confidence intervals for linear regression coefficients;
- 5*) to investigate various types of dependence of the effective attribute on the factor one, selecting the most successful type of regression dependence.

No back.	Assessment criteria	Maximum Points

1.	Written work can be done both individually and in teams of 2-4 people Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; performance of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	10 points
2.	performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	10 points
	Total Points	20 points

Individual task No2

An individual task is performed by a student (or a group of students) and is a set of cases that simulate the practical application of statistical methods.

Case 1. A pharmaceutical company is conducting clinical trials of a new drug to lower cholesterol levels. It is necessary to evaluate the effectiveness of the drug and compare approaches to data analysis (frequentist and Bayesian). The confidence interval and t-statistics in the case of the frequency approach, the confidence interval and the probability of effectiveness in the case of the Bayesian approach are to be evaluated.

Input

Frequency approach

- Experimental group: 200 patients who took a new drug
- Control group: 200 patients taking placebo
- Mean reduction in the experimental group: 5.6 mg/dL
- Standard Deviation: 4.2 mg/dL
- Mean reduction in control: 1.2 mg/dL
- Standard Deviation: 3.8 mg/dL

Bayesian approach

- **A priori mean:** 5 mg/dL (based on similar drug data)
- **Prior variance:** 2.5 (assumed uncertainty)

Case 2. A real estate company collects data on the sale of apartments in a certain area of the city. The goal is to build a predictive model of the price of an apartment based on its characteristics using Bayesian regression. The initial data includes the following parameters:

- The area of the apartment (X1) is in square meters
- Distance to the city center (X2) — in kilometers
- Number of rooms (X3) - Integer
- Presence of an elevator (X4) - binary variable (0 or 1)
- Sale price (Y) is the target variable, the cost of the apartment in thousands of rubles

Requires:

1. Build a Bayesian linear regression model to predict the price of an apartment.
2. Evaluate the a posteriori distribution of model parameters.
3. Make a price forecast for a new apartment with specified characteristics.
4. Assess the uncertainty of the forecast.

Dataset provided (example):

Area	Distance	Rooms	Elevator	Price
60	5	2	1	3500
85	3	3	1	5200
45	7	1	0	2800
95	2	3	1	6000
70	4	2	0	3800

Additional conditions:

- A normal prior distribution is assumed for the coefficients of the model.
- It is necessary to use the MCMC (Markov Chain Monte Carlo) method to evaluate the parameters.
- Compare the results with classical linear regression.

No back.	Assessment criteria	Maximum Points
1.	Written work can be done both individually and in teams of 2-4 people Performance of 90% or more of each task – complete, correct, consistent, competent performance with individual shortcomings; performance of more than 75% and less than 90% of the task – correct execution of tasks with a small number of errors;	5 points
2.	performance of more than 50% and less than 75% of the task – execution of the main part of the tasks with errors; Completion of less than 50% - partial completion of tasks (less than half), making a significant number of mistakes.	10 points
	Total Points	15 points

8.5. Exam Questions and Semester 2 Tickets

List of exam questions

1. Principles of combinatorics. Placements, permutations, combinations.
2. Algebra of events. Operations on events.
3. Axiomatic, classical, geometric, statistical definitions of probability.
4. Independence of events. Theorems of addition, multiplication of events.
5. The formula for full probability. Bayes' formula.
6. A sequence of independent Bernoulli tests. Bernoulli's formula.
7. Approximate formulas in Bernoulli's scheme.
8. The concept of DSV. Methods of setting the DSV. Properties of DSV.
9. The concept of NSV. Methods of setting the NSV. Properties of NSV.
10. Numerical characteristics of the DSV.
11. Numerical characteristics of the NSV.
12. Basic Laws of Distribution of Voluntary Resources.
13. Basic Laws of NSV Distribution.
14. Discrete random vectors. Ways to set them.
15. Continuous random vectors. Ways to set them.
16. Dependent and independent random variables. Conditional Distribution Laws.
17. Numerical characteristics of discrete random vectors.
18. Numerical characteristics of continuous random vectors.
19. The concept of correlation.
20. Random argument functions.
21. Laws of distribution of sum and product of random variables.
22. Numerical characteristics of random argument functions.
23. The law of large numbers.
24. The concept of the central limit theorem.
25. Basic concepts of mathematical statistics. Sampling and ways to describe it.
26. Point estimates of distribution parameters.
27. The method of moments.
28. The method of greatest plausibility.
29. Interval estimates of distribution parameters.
30. Statistical testing of hypotheses.
31. Pearson's criterion of agreement.
32. Parametric and nonparametric criteria.
33. Regression. Tasks of regression analysis.
34. Method of least squares.
35. Checking the adequacy of the regression model.
36. Bayesian regression.
37. The Random Process and Its Characteristics.
38. Markov processes.
39. Hidden Markov processes.
40. Time series.
41. The concept of Bayesian modeling.
42. The law of large numbers.
43. Consequences of the ZBC.
44. Central limit theorem.

8.6. Sample exam ticket (2nd semester)

MINISTRY OF EDUCATION AND SCIENCE OF THE RUSSIAN FEDERATION
Federal State Autonomous Educational
Institution of higher education

"SOUTHERN FEDERAL UNIVERSITY"

EXAM TICKET No 0

In the discipline **SPECIAL SECTIONS OF MATHEMATICS**

Structural unit: **Institute of Computer Technologies and Information Security**

1. The formula for full probability. Bayes' formula.
2. Interval estimates of distribution parameters.
3. From an urn containing 10 balls, 3 of which are black, 2 balls were taken out at random. What is the probability that the balls taken out are black?
4. Find the variance of the DM given density $f(x) = \begin{cases} a(x+1), & -1 < x < 3, \\ 0, & \text{в противном случае.} \end{cases}$
5. Find the corrected sample variance from the sample presented.

x_i	1	2	3	4
n_i	5	3	4	2

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_____ V.V. Duryagina
Head. Department. FMIO _____ G.V. Kupovykh
« » 2025

Evaluation criteria:

35 - 40 points are given to a student if a detailed answer to theoretical questions is given with an appropriate definition, accompanying theorems and consequences, as well as a practical task is correctly solved;

30 - 34 points are given to a student if a correct, but not complete answer to theoretical questions is given, and the practical task is also correctly solved;

20 – 29 points are given to a student when he has answered no more than one theoretical question and has partially coped with the solution of the practical task;

Less than 20 points are given in case of an incorrect answer to both theoretical questions.